Lucas Coefficient Families and Exact Arithmetic for Every Integer Coefficient

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Abstract

We study weighted Lucas expressions for positive integer coefficients. Every positive difference $L_i - 5F_{2r}L_j$, for $r \geq 1$, has a unique representation; we determine its admissible indices, bounded-index count, and asymptotics. In contrast, for every $q = L_{2r}$, both $qL_i - L_j$ and $qL_i + L_j$ have infinitely many multiply represented positive values. We give a complete coefficient classification for infinite collisions in the two reversed-difference and sum forms, including an infinite family of triple representations. A transfer lemma relates sum collisions to reversed-difference collisions. An algebraic-norm bound gives exact representation and rank algorithms for every integer coefficient, and excludes all positive simultaneous-shift collision families for coefficients at least two. This exclusion and effective algebraic-norm estimates yield complete global collision bounds for S_q and a necessary-and-sufficient finite test for infinite collisions in T_q and P_q. The explicit coefficient families decide infinite collision behavior, while finite integer algorithms decide global uniqueness for every coefficient and all three forms.

Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Use $F_{-n} = (-1)^{n+1}F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j : i, j \geq 1\}.$$

The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

The collision-free coefficients begin $5, 15, 40, 105, 275, \ldots$; the persistent-collision coefficients begin $3, 7, 18, 47, 123, \ldots$.

An infinite collision-free family

Theorem 1. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i, j) = (2r + 2, 1) \quad or \quad i \geq 2r + 3,\ 1 \leq j \leq i - 2r - 2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{2}$$

Thus $j = 1$ first becomes admissible at $i = 2r + 2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r + 2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{3}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r + 2$ implies $0 < |i - 4r + 2| \leq 2r - 4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (3), and the same block has distinct values. □

Corollary 2. *For $N \geq 2r + 2$ the number of distinct positive differences with first index at most N equals*

$$1 + \frac{(N - 2r - 2)(N - 2r - 1)}{2}.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and $A_r(x) = \#(S_{5F_{2r}} \cap [1, x])$,

$$A_r(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O_r(\log x), \qquad \log_\varphi b_{r,n} = \sqrt{2n} + O_r(1),$$

where $b_{r,n}$ is the n-th positive value. Its exact value can be recovered by integer binary search.

Proof. Summing block sizes $1, 1, 2, 3, \ldots$ gives the formula. Let $m(x) = \max\{m \geq 1 : L_m \leq x\} = \log_\varphi x + O(1)$. All positive differences with $i \leq m(x)$ are at most x. For $j \geq 2$, $A_2 \geq L_0 = 2$ and $A_3 \geq L_1 = 1$, so recurrence gives $A_j \geq L_{j-2}$. Therefore, for all sufficiently large i, a value in $\mathcal{B}_i$ is at least L_{i-2r-4}; if it is at most x then $i \leq m(x) + 2r + 4$. Applying the triangular count to these two index bounds proves the asymptotic. Inverting it proves the nth-term estimate. An exact count tests $L_j \geq \lceil (L_i - x)/q \rceil$ within finitely many admissible blocks; binary search inverts that monotone count. □

A family with infinitely many collisions

Theorem 3. *For every $r \geq 1$, the coefficient $q = L_{2r}$ has infinitely many repeated positive values in both T_q and P_q. For every $n \geq 1$,*

$$L_{n+4r} = qL_{n+2r} - L_n = qL_{n+6r} - L_{n+8r} > 0, \tag{4}$$

$$qL_{n+2r} + L_{n+8r} = qL_{n+6r} + L_n. \tag{5}$$

The two index pairs in each equality are distinct.

Proof. The product identity $L_aL_b = L_{a+b} + (-1)^a L_{b-a}$ with $a = 2r$ gives

$$qL_{n+2r} = L_{n+4r} + L_n, \quad qL_{n+6r} = L_{n+8r} + L_{n+4r}.$$

These yield (4) and (5). Both resulting sequences of positive values increase strictly with n, providing infinitely many collisions. □

Proposition 4 (Collision transfer)**.** *Suppose S_q has unique positive representations and q divides no Lucas number L_j with $j \geq 1$. Then every sum collision $qL_a + L_b = qL_c + L_d$ with $a < c$ induces a positive collision $qL_a - L_d = qL_c - L_b > 0$ in T_q. These hypotheses hold for every $q = 5F_{2r}$.*

Proof. Set $t = qL_a - L_d = qL_c - L_b$. If $t = 0$, then $q \mid L_d$, a contradiction. If $t < 0$, the equality $L_d - qL_a = L_b - qL_c > 0$ gives two representations in S_q, also a contradiction. Thus $t > 0$. The final assertion follows from Theorem 1 and the four nonzero Lucas residues modulo five. □

Divisibility families of infinite collisions

Theorem 5 (Lucas–Fibonacci quotient coefficients)**.** *Let $m > h \geq 1$ have the same parity, and suppose*

$$q = \begin{cases} L_m/L_h \in \mathbb{Z}, & m, h \text{ odd}, \\ F_m/F_h \in \mathbb{Z}, & m, h \text{ even}. \end{cases}$$

Then $q \geq 2$, and for every $n \geq 1$,

$$qL_{n+m-h} - L_n = qL_{n+m+h} - L_{n+2m} > 0. \tag{6}$$

In particular, T_q and P_q each contain infinitely many positive values having two distinct representations. The coefficient $q = 2$ also has infinitely many collisions in both sets.

Proof. For m odd, the Lucas product formula gives

$$L_{n+2m} - L_n = L_mL_{n+m}, \qquad L_{n+m+h} - L_{n+m-h} = L_hL_{n+m}.$$

For m, h even, the Lucas–Fibonacci identity instead gives

$$L_{n+2m} - L_n = 5F_mF_{n+m}, \quad L_{n+m+h} - L_{n+m-h} = 5F_hF_{n+m}.$$

In either case dividing the two identities gives

$$L_{n+2m} - L_n = q(L_{n+m+h} - L_{n+m-h}),$$

which is (6). The right-hand side of the first expression $qL_{n+m-h} - L_n$ is strictly positive and increases with n because $q \geq 2$ and $m - h \geq 1$. The equivalent sum collision is

$$qL_{n+m-h} + L_{n+2m} = qL_{n+m+h} + L_n.$$

For $q = 2$, the recurrence gives the further family

$$2L_{n+2} - L_n = L_{n+3} = 2L_{n+3} - L_{n+3},$$

again with strictly increasing positive values. The corresponding sum identity follows by swapping the two subtracted indices. □

Remark 6. *The theorem covers* every *Lucas coefficient $q = L_k$ ($k \geq 1$): for odd k choose $(m, h) = (k, 1)$ when $k > 1$, and for even $k = 2r$ choose $(m, h) = (4r, 2r)$ using $F_{4r}/F_{2r} = L_{2r}$. For $q = L_1 = 1$, $L_{n+2} - L_n = L_{n+3} - L_{n+2} > 0$ already provides infinitely many collisions. It also covers infinitely many other integer coefficients, for example $8, 19, 21, 48, 55$.*

Theorem 7 (Half-Lucas collision coefficients)**.** *For every integer $r \geq 0$, let $M = 6r + 3$ and*

$$q = \frac{L_M}{2} \in \mathbb{Z}.$$

Then T_q and P_q each have infinitely many repeated positive values. More precisely, for every $x \geq M$,

$$qL_x - L_{x-M+1} = qL_{x+3} - L_{x+M+1} > 0, \tag{7}$$

and

$$qL_x + L_{x+M+1} = qL_{x+3} + L_{x-M+1}.$$

The coefficients begin $2, 38, 682, 12238, 219602, \ldots$.

Proof. The Lucas residues modulo 2 have period 3, so L_M is even when $3 \mid M$. As M is odd, the product identity gives

$$L_{x+M+1} - L_{x-M+1} = L_M L_{x+1}.$$

The recurrence also gives $L_{x+3} - L_x = 2L_{x+1}$. Substituting $2q = L_M$ proves (7) and the sum identity. As $x \geq M \geq 3$, $qL_x - L_{x-M+1} > 0$; these values strictly increase with x. □

Corollary 8 (Triple collisions for $q = 2$)**.** *For every $n \geq 3$,*

$$\boxed{L_{n+1} = 2L_n - L_{n-2} = 2L_{n+1} - L_{n+1} = 2L_{n+3} - L_{n+4}.}$$

Thus T_2 has infinitely many values with at least three distinct representations.

Proof. The first equality follows from $L_{n+1} = 2L_n - L_{n-2}$, the second is immediate, and the third is Theorem 7 with $r = 0$. □

Complete classification of stationary collisions

The quotient and half-Lucas constructions above in fact exhaust all coefficients admitting infinitely many positive collisions in T_q and P_q. We first classify every possible pair of index gaps.

Lemma 9 (Rigidity of the two index gaps)**.** *Suppose $q \geq 2$, $i < k$, $j < \ell$, and*

$$q(L_{n+k} - L_{n+i}) = L_{n+\ell} - L_{n+j} \qquad (n \geq 0). \tag{8}$$

Set $h = k - i$, $e = \ell - j$, and $d = i - j$. The only possibilities are:

1. *$h = 2r$, $e = 2m$, $m > r \geq 1$, with $m \equiv r \pmod 2$, $d = m - r$, and either $q = L_m/L_r$ for m, r odd or $q = F_m/F_r$ for m, r even;*
2. *$h = 1$, $e = 2m$, $m \geq 3$ odd, $d = m + 1$, and $q = L_m$;*
3. *$h = 3$, $e = 2m$, $m \geq 3$ odd and divisible by 3, $d = m - 1$, and $q = L_m/2$;*
4. *$(h, e, d, q) = (1, 3, 2, 2)$ or $(2, 3, 0, 2)$.*

Every identity on this list is realized, with positive values and arbitrarily large simultaneous shifts.

Proof. Let $\alpha = \varphi$, $\beta = -\alpha^{-1}$, and set

$$C(s) = L_s - 1 - (-1)^s, \qquad \Phi(s) = \frac{1+\alpha^{-s}}{1-\alpha^{-s}} \quad (s \geq 1).$$

The two conjugate Binet coefficient equations of (8) give

$$q\alpha^i(\alpha^h - 1) = \alpha^j(\alpha^e - 1), \qquad q\beta^i(\beta^h - 1) = \beta^j(\beta^e - 1).$$

Their norms yield

$$q^2 C(h) = C(e), \qquad i \equiv j \pmod 2. \tag{9}$$

In particular $e > h$, and $d = i - j$ is even. Dividing the two Binet equations, and using $\beta = -\alpha^{-1}$, gives the exact relation

$$\alpha^{2d} = \alpha^{e-h} \frac{\Phi(h)^{[h \text{ odd}]}}{\Phi(e)^{[e \text{ odd}]}}. \tag{10}$$

The function $\Phi(s)$ decreases strictly to 1; direct calculation gives

$$\Phi(1) = \alpha^3, \qquad \Phi(3) = \alpha. \tag{11}$$

If h, e are both even, (10) gives $2d = e - h$. Write $h = 2r$, $e = 2m$. Since d is even, $m \equiv r \pmod 2$. For even indices,

$$C(2s) = L_{2s} - 2 = \begin{cases} L_s^2, & s \text{ odd}, \\ 5F_s^2, & s \text{ even}. \end{cases}$$

Equation (9) gives the two quotient conditions in case 1.

If both gaps are odd, put $u = (e-h)/2$. Equation (10) becomes

$$\alpha^{2(d-u)} = \Phi(h)/\Phi(e).$$

When $h \geq 3$ this ratio lies strictly between 1 and $\Phi(3) = \alpha < \alpha^2$, which is impossible for an integral exponent $2(d-u)$. When $h = 1$, the ratio is at least $\Phi(1)/\Phi(3) = \alpha^2$, with equality at $e = 3$; for $e > 3$ it lies strictly between α^2 and $\alpha^3 < \alpha^4$. Therefore $h = 1, e = 3$; then $d = 2$ and (9) gives $q = 2$.

If h is odd and e is even, the extra factor in (10) is $\Phi(h)$. For odd $h \geq 5$, the right side lies strictly between consecutive powers α^{e-h} and α^{e-h+1}, of which only the second has an even exponent. This is impossible. For $h = 1$, (11) gives $2d = e + 2$; for $h = 3$ it gives $2d = e - 2$. Putting $e = 2m$ in the norm equation then gives, respectively, $q = L_m$ or $q = L_m/2$ when m is odd. Indeed m even would require an irrational positive multiple of $\sqrt{5}$ to be the integer q. In the second case, integrality of $L_m/2$ is equivalent to $3 \mid m$; the condition $q \geq 2$ also excludes $m = 1$.

Finally, for even h, odd e, (10) has the factor $1/\Phi(e)$. For $e \geq 5$, the exponent lies strictly between the consecutive integers $e - h - 1$ and $e - h$; the former is even, and no even exponent is possible. Thus $e = 3$, necessarily $h = 2$, and (11) gives $d = 0$ and $q = 2$.

For sufficiency, case 1 is the quotient identity (6). Cases 2 and 3 follow from $L_{n+2m} - L_n = L_m L_{n+m}$ for odd m, together with $L_{n+1} - L_n = L_{n-1}$ and $L_{n+3} - L_n = 2L_{n+1}$. The last two cases are the corresponding $q = 2$ identities. All hold for every sufficiently large common shift, with strictly increasing positive values. □

Theorem 10 (Explicit infinite-collision coefficient classification)**.** *For every integer $q \geq 2$, the set T_q has infinitely many multiply represented positive values if and only if q belongs to the following union:*

$$\boxed{\begin{aligned} &\{L_m/L_r \in \mathbb{Z} : m > r \geq 1,\ m, r \text{ odd}\} \\ \cup\ &\{F_m/F_r \in \mathbb{Z} : m > r \geq 2,\ m, r \text{ even}\} \\ \cup\ &\{L_{6r+3}/2 : r \geq 0\}. \end{aligned}} \tag{12}$$

The same equivalence holds for P_q. Every other integer $q \geq 2$ has only finitely many collisions in T_q and P_q.

Proof. Each coefficient in (12) has an infinite family of positive collision values by Theorems 5 and 7. Conversely, if T_q has infinitely many collision values, Theorem 16 forces a collision holding under every common index shift. Lemma 9 puts its coefficient in one of the three families: cases 2 and 4 are subsumed by the first or third family, and case 3 is the third family. The equivalence for P_q and finiteness outside these families follow from Corollary 17. □

An exact algorithm for every integer coefficient

The coefficient families above admit efficient special-purpose rankings. An integer-bound argument gives complete, exact arithmetic for every coefficient, with no conjectural collision classification needed.

Theorem 11 (Uniform finite search bounds)**.** *Let $q \geq 1$ and $X \geq 1$ be integers, and put*

$$B = (q+1)(X+q+1)+1.$$

For any $0 < L_i - qL_j \leq X$ with $i, j \geq 1$, necessarily

$$L_j \leq B, \qquad L_i \leq X + qB. \tag{13}$$

For $q \geq 2$ and $0 < qL_i - L_j \leq X$, necessarily

$$L_i \leq M := \max\{B, \lfloor X/(q-1) \rfloor\}, \qquad L_j \leq qM. \tag{14}$$

When $q = 1$, the expressions $qL_i - L_j$ and $L_i - qL_j$ coincide and obey (13). For $q \geq 1$ and $0 < qL_i + L_j \leq X$,

$$L_i \leq \lfloor (X-1)/q \rfloor, \qquad L_j \leq X - q. \tag{15}$$

Consequently, for every *$q \in \mathbb{Z}$, the set of distinct positive values at most X in each of S_q, T_q, P_q is effectively computable by finite integer arithmetic. For every nonzero q, all representations of each positive value can also be enumerated, yielding its exact multiplicity. For $q = 0$, the positive sets S_0 and P_0 are the positive-index Lucas numbers, each with infinitely many representations; T_0 is empty.*

Proof. Write $\alpha = \varphi$ and $\beta = (1-\sqrt{5})/2 = -1/\varphi$. For $d \geq 1$ and $q \geq 1$, the algebraic integer $\alpha^d - q$ has nonzero integral norm

$$(\alpha^d - q)(\beta^d - q) \in \mathbb{Z} \setminus \{0\}.$$

It cannot vanish because $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q + 1$, this gives

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{16}$$

A positive difference $L_i - qL_j$ has $i > j$; write $d = i - j$. By Binet's formula,

$$L_i - qL_j = \alpha^j(\alpha^d - q) + \beta^i - q\beta^j.$$

For $0 < L_i - qL_j \le X$, it follows that

$$\frac{\alpha^j}{q+1} \le X + |\beta^i - q\beta^j| \le X + q + 1.$$

Thus $\alpha^j \le (q+1)(X+q+1)$, and $L_j = \alpha^j + \beta^j \le B$. The equality $L_i = x + qL_j$ now proves (13).

For $0 < qL_i - L_j \le X$, when $j > i$, the same argument with $d = j - i$ shows $L_i \le B$, hence $L_j = qL_i - x \le qB$. When $j \le i$ and $q \ge 2$, $x = qL_i - L_j \ge (q-1)L_i$; this proves (14). The sum bounds (15) follow from $L_i, L_j \ge 1$.

For negative coefficients, write $m = -q > 0$. Then $S_{-m} = P_m$ with the indices interchanged, $P_{-m} = S_m$ with the indices interchanged, and T_{-m} is empty. The assertions for $q = 0$ are immediate. Thus the finite index bounds cover all integers. □

Corollary 12 (Universal exact counting and unranking)**.** *For any integer q, expression type S, T or P, and integer $X \ge 0$, there is a terminating algorithm for the number of* distinct *positive values at most X, exact membership, and all positive representations (or detection of infinite multiplicity when $q = 0$). Whenever the positive set is nonempty, its n-th distinct value is computable exactly for every $n \ge 1$.*

Proof. Enumerate the finitely many indices specified by Theorem 11, retain the positive values at most X, and deduplicate integer values. Each nonempty positive set is unbounded: for S_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, for T_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, and for P_q with $q > 0$ fix $i = 1$ and let $j \to \infty$. The negative and zero cases reduce to these or the Lucas numbers themselves. Therefore doubling the threshold until n values are found and then binary-searching the exact count always terminates. The only empty case is T_q for $q \le 0$. □

Proposition 13 (No simultaneous-shift positive collisions)**.** *For every integer $q \ge 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \ge 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \ge 1$, $e = \ell - j \ge 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \ge 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \ge 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \le \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \ge 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

Finite global collisions for every coefficient $q \ge 2$

A finite threshold is sufficient to compute any term. The stronger statement below provides a bound independent of the value.

Theorem 14 (Uniform complete collision certificate)**.** *Fix an integer $q \geq 2$. Let $t = t(q)$ be the least positive index such that*

$$L_t > 200(q+1)^4. \tag{17}$$

If two distinct positive representations satisfy

$$L_i - qL_j = L_k - qL_\ell > 0, \qquad i < k,$$

then

$$\boxed{k \leq 6t.} \tag{18}$$

Consequently every S_q with $q \geq 2$ has only finitely many multiply represented positive values. Its complete collision list and global uniqueness can be determined by checking the finitely many pairs $1 \leq j < i \leq 6t$.

Proof. Write $\alpha = \varphi$, $\beta = -\alpha^{-1}$. The equality forces $j < \ell$. Moreover, positivity gives $k > \ell$. Put $d = k - \ell$, $h = k - i$ and $g = k - j$. We prove $k \leq 6t$ by contradiction. The bounds

$$\frac{1}{2}\alpha^n \leq L_n \leq 2\alpha^n \quad (n \geq 0), \qquad L_k - L_i \geq L_{k-2}$$

follow from Binet's formula and monotonicity. Since

$$L_{k-2} \leq L_k - L_i = q(L_\ell - L_j) < qL_\ell,$$

we have $\alpha^d < 4q\alpha^2 < \alpha^t$; thus $1 \leq d < t$. Suppose $k > 6t$. Then $\ell = k - d > 5t$.

Write

$$x = L_k - qL_\ell = \alpha^\ell(\alpha^d - q) + \beta^k - q\beta^\ell.$$

The norm argument in (16) gives $|\alpha^d - q| \geq 1/(q+1)$. Since $|\beta^k - q\beta^\ell| \leq q+1$ and $\alpha^\ell > 2(q+1)^2$, the positivity of x forces $\alpha^d - q > 0$ and

$$x \geq \frac{\alpha^\ell}{2(q+1)}.$$

But $x < L_i \leq 2\alpha^i$, so $\alpha^{\ell-i} < 4(q+1) < \alpha^t$, which gives $1 \leq h < 2t$.

Next, the original equality can be rearranged as

$$qL_j = L_i - L_k + qL_\ell = \alpha^k A + \beta^k A',$$

where

$$A = \alpha^{-h} - 1 + q\alpha^{-d}, \qquad A' = \beta^{-h} - 1 + q\beta^{-d}.$$

Both are conjugate algebraic integers, and

$$|A'| \leq \alpha^h + 1 + q\alpha^d < 3\alpha^{2t} < \alpha^{3t}.$$

If $A = 0$, then $qL_j = |\beta^k A'| < 1$, an impossibility. Otherwise AA' is a nonzero integer, so $|A| \geq \alpha^{-3t}$. As $k > 6t$ and $qL_j > 0$, this forces $A > 0$ and

$$qL_j > \tfrac{1}{2}\alpha^{k-3t}.$$

Using $qL_j \leq 2q\alpha^j$ gives $\alpha^{k-j} < 4q\alpha^{3t} < \alpha^{4t}$, hence $g < 4t$.

Finally, express the collision as

$$0 = L_k - L_i - qL_\ell + qL_j = \alpha^k E + \beta^k E',$$

with the conjugate algebraic integers

$$E = 1 - \alpha^{-h} - q\alpha^{-d} + q\alpha^{-g}, \quad E' = 1 - \beta^{-h} - q\beta^{-d} + q\beta^{-g}.$$

The bounds on d, h, g imply

$$|E'| \le 1 + \alpha^{2t} + q\alpha^{t} + q\alpha^{4t} < \alpha^{6t}.$$

If $E \neq 0$, the integral norm $EE' \neq 0$ gives $|E| \ge 1/|E'|$. The displayed equality now implies $\alpha^{2k} \le |E'|^2 < \alpha^{12t}$, contrary to $k > 6t$.

If $E = 0$, the equality also forces $E' = 0$. It then holds after every simultaneous translation of all four indices. Because $\alpha^d - q > 0$, the translated differences are positive for all sufficiently large translations, contradicting Proposition 13. Thus neither case is possible, and $k \le 6t$. □

Corollary 15 (Complete coefficient-by-coefficient uniqueness decisions and growth)**.** *For any input integer $q \ge 2$, a finite, exact integer algorithm decides whether every positive value $L_i - qL_j$ is uniquely represented. If not, it returns every multiply represented value together with all its index pairs. Furthermore, for each fixed $q \ge 2$,*

$$\#\{x \le X : x \in S_q\} = \tfrac{1}{2}(\log_\varphi X)^2 + O_q(\log X).$$

Proof. All first indices of a collision are at most $6t$ by Theorem 14; finite enumeration therefore gives a complete answer. For the asymptotic, choose an integer d with $\alpha^{d-1} < q < \alpha^d$. The Binet formula shows that, for all sufficiently large j, the sign of $L_{j+s} - qL_j$ is negative if $s < d$ and positive if $s \ge d$. Hence the number of positive index pairs with first index at most N is $\frac{1}{2}N^2 + O_q(N)$. Those with $L_i \le X$ are automatically at most X. Conversely, (13) implies $L_i \le X + q((q+1)(X+q+1)+1) = O_q(X)$ for every positive representation at most X. Thus all contributing first indices satisfy $i \le \log_\varphi X + O_q(1)$. Theorem 14 removes only a finite number of repeated values, giving the result. □

Complete effective classification of infinite collisions

Theorem 16 (Finite stationary pattern criterion)**.** *Fix any integer $q \ge 2$ and choose t as in (17). If*

$$qL_k - L_\ell = qL_i - L_j > 0, \qquad i < k, \tag{19}$$

and $k > 12t$, then all four indices lie within $6t$ of k, apart from a possible upper extension of at most t, and the same equality holds after every *simultaneous nonnegative shift of its indices. Consequently, the set T_q has infinitely many multiply represented positive values if and only if there are indices with*

$$K = 12t + 1, \quad K - 3t \le i < K, \quad K - t \le \ell \le K + t, \quad K - 6t \le j < \ell$$

such that

$$0 < qL_K - L_\ell = qL_i - L_j, \qquad qL_{K+1} - L_{\ell+1} = qL_{i+1} - L_{j+1}. \tag{20}$$

This is a finite, exact, necessary-and-sufficient integer test for the existence of infinitely many positive collisions in T_q. Every collision not belonging to a stationary family has larger first index $k \le 12t$.

Proof. Equation (19) forces $j < \ell$ and $q(L_k - L_i) = L_\ell - L_j$. The bounds $L_{n-2} \le L_n - L_{n-1}$ and $\frac{1}{2}\alpha^n \le L_n \le 2\alpha^n$ yield

$$qL_{k-2} \le L_\ell, \qquad L_{\ell-2} \le qL_k.$$

Consequently, with $\delta = \ell - k$,

$$\alpha^{-\delta} < 4\alpha^2/q < \alpha^t, \qquad \alpha^\delta < 4q\alpha^2 < \alpha^t,$$

so $|\delta| < t$. Assume $k > 12t$. Set

$$x = qL_k - L_\ell = \alpha^k A + \beta^k A', \quad A = q - \alpha^\delta, \quad A' = q - \beta^\delta.$$

The conjugate algebraic integers A, A' have nonzero integral norm, including when $\delta = 0$ (since then $A = q - 1$). As $|A'| < \alpha^{2t}$, we have $|A| > \alpha^{-2t}$. Positivity of x and $k > 12t$ imply $A > 0$ and $x > \frac{1}{2}\alpha^{k-2t}$. Since $x < qL_i \le 2q\alpha^i$, the gap $h = k - i$ satisfies $0 < h < 3t$.

Next,

$$L_j = qL_i - qL_k + L_\ell = \alpha^k B + \beta^k B',$$

where $B = q\alpha^{-h} - q + \alpha^\delta$ and $B' = q\beta^{-h} - q + \beta^\delta$. The conjugates are algebraic integers and $|B'| < \alpha^{5t}$. If $B = 0$, then $L_j < 1$, a contradiction; otherwise their integral norm gives $|B| > \alpha^{-5t}$. As $k > 12t$ and $L_j > 0$, we obtain $L_j > \frac{1}{2}\alpha^{k-5t}$. Thus $g = k - j < 6t$. Since $j < \ell$, also $g > -t$.

Now the collision is

$$0 = qL_k - qL_i - L_\ell + L_j = \alpha^k E + \beta^k E',$$

where

$$E = q - q\alpha^{-h} - \alpha^\delta + \alpha^{-g}, \quad E' = q - q\beta^{-h} - \beta^\delta + \beta^{-g}.$$

Both are algebraic integers and $|E'| < \alpha^{8t}$. If $E \neq 0$, the integral norm gives $|E| > 1/|E'|$ and the equality forces $\alpha^k \le |E'| < \alpha^{8t}$, contradicting $k > 12t$. Therefore $E = E' = 0$, and the identity holds under all simultaneous shifts. The positive leading coefficient A makes its common values positive and unbounded under increasing shifts.

Every infinite sequence of distinct collision values must have unbounded k, and so contains a stationary identity of the preceding form. Translate its indices to $k = K = 12t + 1$. The established gap bounds give precisely the finite ranges in (20). The leading coefficient $A > 0$ ensures positivity at K. Conversely, the two equalities in (20) give consecutive zeros of a Lucas-recursive difference. They therefore give the identity at all shifts, and its positive values become arbitrarily large. This proves the criterion and the asserted bound for nonstationary collisions. □

Corollary 17 (All integer signs and all three forms)**.** *For any integer q, it is decidable by finite exact arithmetic whether S_q, T_q, or P_q has infinitely many multiply represented positive values. For $q \ge 2$, S_q always has only finitely many collisions, while T_q and P_q have infinitely many collisions* simultaneously *or neither does. In the finite case the exact collision list is computable; in the infinite case all collisions are covered by finitely many small-index exceptions and finitely many stationary families. Negative and zero coefficients reduce to the corresponding positive cases or elementary Lucas sets.*

Proof. For $q \ge 2$, Theorem 14 settles S_q, and Theorem 16 settles T_q, including its finite exceptions. Every positive collision in T_q gives a collision in P_q by exchanging the subtracted second indices. Conversely, a collision in P_q,

$$qL_k + L_\ell = qL_i + L_j, \qquad i < k, \quad \ell < j,$$

gives $v = qL_k - L_j = qL_i - L_\ell$. If $v > 0$ it is a collision in T_q. If $v < 0$, it is a positive collision in S_q, and only finitely many such index quadruples exist by Theorem 14. If $v = 0$, both $qL_k = L_j$ and $qL_i = L_\ell$ hold. The norm bound (16), applied at value zero, bounds their indices for fixed q. Thus P_q has infinitely many collisions exactly when T_q does. For $q = 1$, the Lucas recurrence gives infinitely many collisions in all three sets. For $q = 0$, T_0 is empty, while S_0 and P_0 are Lucas sets with infinite representation multiplicities. For $q = -m < 0$, T_q is empty, $S_q = P_m$ and $P_q = S_m$ after exchanging the index roles. All decisions therefore reduce to the preceding finite algorithms. The finite exceptional index combinations can be enumerated using the same bounds. □

Corollary 18 (Complete global uniqueness decision for all integer q)**.** *For every $q \in \mathbb{Z}$ and each of the three forms S_q, T_q, P_q, exact finite integer arithmetic determines whether all positive values have unique index-pair representations. For $q \geq 2$, the procedure is:*

1. *For S_q, enumerate indices up to $6t$, as in Theorem 14.*
2. *For T_q or P_q, first perform the finite stationary-pattern test (20). If a stationary pattern exists, uniqueness fails with infinitely many collisions. Otherwise, enumerate all index pairs $1 \leq i, j \leq 13t$ and check for repeated positive values.*

The cases $q \leq 1$ reduce to the preceding cases by sign exchange, or are explicit empty or infinitely repeated Lucas sets.

Proof. The first assertion for S_q is Theorem 14. Suppose that $q \geq 2$ has no stationary collision. Every T_q collision then has larger first index at most $12t$ by Theorem 16. The other first index is smaller, while each second index of a positive value satisfies $L_j < qL_i$ and hence $j < i + t$. Thus all its indices are at most $13t$.

For a P_q collision with first indices $i < k$, write $v = qL_k - L_j = qL_i - L_\ell$, as in the proof of Corollary 17. If $v > 0$, the corresponding T_q collision has all indices at most $13t$. If $v < 0$, the corresponding S_q collision has all indices at most $6t$. If $v = 0$, the equation $qL_a = L_b$ implies

$$\alpha^a |\alpha^{b-a} - q| \leq |\beta^b - q\beta^a| \leq q + 1.$$

The norm bound (16) gives $\alpha^a \leq (q+1)^2$; since $L_b = qL_a$, also $\alpha^b < \alpha^t$. Thus these index pairs lie below t. All possible P_q collisions therefore occur inside the stated $13t$ window. Negative and small coefficients follow from Corollary 17. □

Context. The case $q = 5F_2 = 5$ generalizes the unique representation of positive differences in OEIS A191981. The quotient and half-Lucas families exhaust all coefficients with infinite collisions in the reversed-difference and sum forms; finite algorithms decide global uniqueness in every form.